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Integrated control of horizontal and vertical movements of a long-stroke pumping unit

Abstract

A long-stroke pumping unit can be moved away from and toward the wellhead by using a single hydraulic power source via a control interface that is coupled to the hydraulic power source and configured to control both the vertical movement and the horizontal movement of the long-stroke pumping unit. The control interface can control the hydraulic power source for extension and retraction of a hydraulic cylinder for horizontal movement of the unit, and the same control interface can control the same hydraulic power source for extension and retraction of a first vertical hydraulic jack and a second vertical hydraulic jack of the long-stroke pumping unit for vertical movement of an end of the long-stroke pumping unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional patent application claiming the benefit of, and priority to, U.S. Provisional Patent Application No. 63/585,798, filed Sep. 27, 2023, which is incorporated by reference herein in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to long-stroke pumping units, and more particularly relates to moving the long-stroke pumping unit away from the wellhead, for example, to service or repair equipment at the wellhead or in the wellbore.

BACKGROUND

(2) A wellbore can be drilled into a subterranean formation to produce hydrocarbons from a producing portion of the subterranean formation. An artificial lift system, such as a reciprocating rod pumping unit, can be used to carry a production fluid (e.g., containing hydrocarbon fluid) from the subterranean formation, through the wellbore, and to a wellhead located at a surface of the earth. The reciprocating rod pumping unit can include a long-stroke pumping unit, a rod string, and a downhole pump. The reciprocating rod pumping unit has an upstroke and a downstroke that pumps the wellbore fluids from the wellbore to the wellhead.

(3) The long-stroke pumping unit is the surface equipment that is located next to the wellhead. At times, access to the wellhead is needed, such as for repair or maintenance of equipment at the wellhead, of the rod string, of the downhole pump, or combinations thereof. However, portions of the long-stroke pumping unit (e.g., a tower of a tower-style unit) can be positioned on the surface of the earth next to the wellhead such that access to the wellhead by humans is obstructed. In some scenarios, the long-stroke pumping unit is moved in a direction away from the wellhead to create more space between the long-stroke pumping unit and the wellhead, facilitating access to the wellhead for the repair or maintenance. When access to the wellhead is no longer needed, the long-stroke pumping unit can be moved back into position for operation.

(4) Due to the large size and height of a tower-style long-stroke pumping unit, moving the unit can be complex and dangerous. For example, wheels can be added to the base of the tower-style long-stroke pumping unit to move the long-stroke pumping unit away from and toward the wellhead; however, the tower can be leaned to the side to add and remove the wheels. Leaning the tower risks tipping over the long-stroke pumping unit. Moreover, moving the unit involves multiple field personnel to provide the lifting equipment, attach the lifting equipment to the unit, move the unit away from the wellhead, move the unit toward the wellhead (after access to the wellhead is no longer needed), and disengage the lifting equipment from the unit.

SUMMARY

(5) In some aspects, the techniques described herein relate to a long-stroke pumping unit including: a base frame including a pair of parallel rails, wherein the base frame is configured to be positioned on a foundation that is proximate to a wellhead, wherein an end of the base frame faces the wellhead; a tower connected to each of the pair of parallel rails near the end of the base frame; a hydraulic cylinder i) having an end connected to the base frame or to the tower and ii) an opposite end connected to the foundation at a location that is between the pair of parallel rails; a first vertical hydraulic jack coupled to one of the pair of parallel rails at the end of the base frame; a second vertical hydraulic jack coupled to another of the pair of parallel rails at the end of the base frame; a hydraulic power source fluidly connected to the hydraulic cylinder, to the first vertical hydraulic jack, and to the second vertical hydraulic jack; and a control interface coupled to the hydraulic power source and configured to control the hydraulic power source for a movement of i) the hydraulic cylinder between a first extended position and a first retracted position, ii) the first vertical hydraulic jack between a second extended position and a second retracted position, and iii)

the second vertical hydraulic jack between a third extended position and a third retracted position.

(6) In some aspects, the techniques described herein relate to a method including: simultaneously actuating, with a hydraulic power source when a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the long-stroke pumping unit; placing wheels onto the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit while the wheels are on the long-stroke pumping unit.

(7) In some aspects, the techniques described herein relate to a method including: simultaneously actuating, with a hydraulic power source while a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit while the long-stroke pumping unit is in a first position; removing wheels from the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source while the hydraulic cylinder of the long-stroke pumping unit is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

(8) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

- (2) FIG. 1A illustrates a side elevational view of a long-stroke pumping unit in a first position.
- (3) FIG. 1B illustrates a side elevational view of the long-stroke pumping unit in a second position.
- (4) FIG. 2A illustrates a top view of the long-stroke pumping unit in the first position, with the hydraulic cylinder in an extended position.
- (5) FIG. 2B illustrates a top view of the long-stroke pumping unit in the second position, with the hydraulic cylinder in a retracted position.
- (6) FIG. 3 illustrates an end elevational view of the long-stroke pumping unit.
- (7) FIG. 4A illustrates a close-up end elevational view of the long-stroke pumping unit, with the vertical hydraulic jacks in a retracted position.
- (8) FIG. 4B illustrates a close-up end elevational view of the long-stroke pumping unit, with the vertical hydraulic jacks in an extended position.
- (9) FIG. 5 illustrates a perspective view of a vertical hydraulic jack on the end of the long-stroke pumping unit.

DETAILED DESCRIPTION

(10) Disclosed is a long-stroke pumping unit that can be moved away from and toward the wellhead without leaning the tower to the side and with fewer field personnel. The movement of the long-stroke pumping unit can be controlled using a single hydraulic power source via a control interface that is coupled to the hydraulic power source and configured to control both the vertical movement and the horizontal movement of the long-stroke pumping unit. The control interface can control the hydraulic power source for extension and retraction of a hydraulic cylinder for

horizontal movement of the unit, and the same control interface can control the same hydraulic power source for extension and retraction of a first vertical hydraulic jack and a second vertical hydraulic jack of the long-stroke pumping unit for vertical movement of an end of the long-stroke pumping unit. The control interface is a single point of control for the movement of the long-stroke pumping unit, and the lifting and moving equipment (lifting jacks for vertical movement and hydraulic cylinder for horizontal movement) are integrated with the long-stroke pumping unit. Single point control and equipment integration can reduce the number of field personnel needed to move the long-stroke pumping unit for access to the wellhead and return of the unit for resuming operation. The long-stroke pumping unit described herein is a tower-style long-stroke pumping unit; however, it is contemplated that the vertical and horizontal movement equipment, configurations, and methods disclosed herein can be applied to other types of long-stroke pumping units.

(11) FIG. 1A illustrates a side elevational view of a long-stroke pumping unit **100** in a first position, and FIG. 1B illustrates a side elevational view of a long-stroke pumping unit **100** in a second position. In the first position, the long-stroke pumping unit **100** is positioned on a foundation **10** next to a wellhead **200** at the surface **20** of the earth. In the second position, the long-stroke pumping unit **100** is moved horizontally away in the direction of double-headed arrow A from the wellhead **200** such that a distance between the long-stroke pumping unit **100** and the wellhead **200** when the long-stroke pumping unit **100** is in the second position is greater than a distance between the long-stroke pumping unit **100** and the wellhead **200** when the long-stroke pumping unit **100** is in the first position.

(12) In both FIGS. 1A and 1B, a wellbore **201** fluidly connects the wellhead **200** with a producing zone of the subterranean formation **21** in which the wellbore **201** is formed. Casing **202** may extend from the wellhead **200** into at least a portion of the wellbore **201**, for example, cemented to the wall of the wellbore **201**. Production tubing **203** can extend from the wellhead **200** and into the wellbore **201** within the casing **202**. The production tubing **203** can be fluidly connected to a downhole pump.

(13) The wellhead **200** can include any equipment known in the art with the aid of this disclosure, such as a production tree, stuffing box, seals, or combinations thereof. The wellhead **200** fluidly connects with a hydrocarbon production line **204**, through which produced fluid flows from the wellhead **200** to another location, such as a storage vessel or pipeline. A polished rod **205** extends through the wellhead **200** (e.g., via seals to prevent leakage of produced fluid from the wellhead **200**) and is connected to a rod string **206**. The rod string **206** is connected to a plunger **207** that travels upward and downward in the production tubing **203** to move fluids into the hydrocarbon production line **204**. The polished rod **205** is coupled to the long-stroke pumping unit **100**.

(14) In aspects, the long-stroke pumping unit **100** can be embodied as any tower-type long-stroke pumping unit known in the art with the aid of this disclosure. For example, the long-stroke pumping unit **100** can include a base frame **101**, a tower **102** positioned on an end **103a** of the base frame **101**, a prime mover **104** coupled to equipment in the tower **102** via a shaft **105** and positioned on the base frame **101**, a control interface **106** for controlling the mechanical equipment in the long-stroke pumping unit **100**, a hydraulic power source **107** positioned on the base frame **101** and coupled to the hydraulic cylinder and vertical hydraulic jacks disclosed herein, and a control interface **108** coupled to the hydraulic power source **107** and configured to control the hydraulic power source **107** for extension and retraction of the hydraulic cylinder and vertical hydraulic jacks (for movement of the long-stroke pumping unit **100** vertically and horizontally).

(15) The foundation **10** can be an immovable, level structure, such as a foundation formed of concrete or cement. The foundation **10** can be a single structure formed to support the base frame **101** of the long-stroke pumping unit **100**. Alternatively, for some embodiments of the base frame **101**, the foundation **10** can be multiple structures that are formed to support the base frame **101**. The foundation **10** is generally configured such that the base frame **101** can be moved horizontally

in the direction of double headed arrow A-A with respect to the foundation **10**.

(16) The base frame **101** is a structure that supports the tower **102**, prime mover **104**, control interface **106**, hydraulic power source **107**, and control interface **108**. The base frame **101** sits or rests on the foundation **10** and is configured to be positioned on the foundation **10** proximate to the wellhead **200** (at a well site). In aspects, the base frame **101** is not connected to the foundation **10** and is horizontally movable relative to the foundation **10** in the direction of double-headed arrow A-A. In some aspects, the base frame **101** can include a pair of parallel rails (see FIG. **1B**, rails **101a** and **101b**). The pair of parallel rails can be connected to one another via one or more cross-members (see FIG. **2A**, cross members **101c** and **101d**). For example, cross-members can connect to each of the rails at end **103a** and opposite end **103b** of the base frame **101**; additionally or alternatively, cross-members can connect to each of the rails at any location along the rails, for example, to support equipment such as the prime mover **104** and the hydraulic power source **107**. In aspects, the parallel rails and the cross-member(s) of the base frame **101** can be formed of interconnected carbon steel beams, for example. The base frame **101** can also be referred to as a skid.

(17) The tower **102** of long-stroke pumping unit **100** can include a housing **109**, a drive sprocket **110**, a chain **111**, a chain idler **112**, a carriage **113**, a counterweight assembly **114**, a top **115**, a drum assembly **116**, a braking system **117**, a load belt **118**, and a hanger assembly **119**. The configuration of the tower **102** is by example only, and other configurations of the tower **102** are contemplated to fall within the scope of this disclosure.

(18) The housing **109** can be a metal structure configured to house, enclose, and/or support the drive sprocket **110**, the chain **111**, the chain idler **112**, the carriage **113**, the counterweight assembly **114**, the top **115**, the drum assembly **116**, the braking system **117**, and the load belt **118**. The base of the housing **109** can include an amount of lubricant for lubricating the chain **111** as the chain **111** is rotated around the drive sprocket **110**. The drive sprocket **110** is mechanically coupled to the prime mover **104** via the shaft **105**, and the drive sprocket **110** is also coupled to the chain **111**. The chain **111** is also coupled to the carriage **113**. In aspects, the chain **111** additionally can be coupled with a chain idler **112** that can be mounted to the housing **109** and configured to maintain a tension of the chain **111** to a setpoint tension. The carriage **113** can be connected to the chain **111** and to the counterweight assembly **114**. In aspects, the carriage **113** is configured to allow a transverse movement of the chain **111** relative to the counterweight assembly **114**. The counterweight assembly **114** is movable up and down within the housing **109** of the tower **102**. The counterweight assembly **114** can include a weight box, one or more counterweights disposed in the weight box, and guide wheels configured to guide the weight box as the counterweight assembly **114** moves in the housing **109**. The guide wheels may be connected to the weight box such that the wheels roll along the inside of the housing **109** as the counterweight assembly **114** moves upward and downward within the housing **109**. In aspects, a weight of the counterweights can correspond to the weight of the rod string **206** and the weight of the fluid produced in a single stroke of the long-stroke pumping unit **100**, such as being equal to a sum of the weight of the rod string **206** and one-half the weight of the fluid produced in a single stroke. The top **115** can be a frame structure that defines the top of the tower **102**. The drum assembly **116** is coupled to the top **115** and can include a drum, a shaft, one or more ribs connecting the drum to the shaft, one or more pillow blocks mounted to the top **115**, and one or more bearings configured to support the shaft while facilitating rotation of the shaft relative to the pillow blocks. The braking system **117** can include one or more disk brakes or drum brakes, that in aspects, can be operated with hydraulic fluid from the hydraulic power source **107** (e.g., the control interface **108** can be additionally configured to control a supply of pressurized hydraulic fluid to the braking system **117**). Alternatively, the braking system **117** can be pneumatically operated. The load belt **118** is a wide and flat belt that has a first end connected to a top of the weight box of the counterweight assembly **114** and a second end coupled to polished rod **205** of the wellhead **200** (e.g., via a hanger assembly **119**). The load belt **118** can extend from

the top of the counterweight assembly **114** upward through the housing **109** of the tower **102** and upward through the top **115**, over an outer surface of the drum of the drum assembly **116**, and downward from the drum assembly **116** to the hanger assembly **119**. The hanger assembly **119** is connected to the polished rod **205** and to the load belt **118**. An optional load sensor can be included between the hanger assembly **119** and the polished rod **205** and configured to send a signal indicating a tension of the rod string **206** to the control interface **106**.

(19) In some aspects, the prime mover **104** can include an electric motor powered with electricity produced from a generator (powered by diesel or other hydrocarbon) or obtained from an electrical grid, or the prime mover **104** can include an internal combustion engine fueled by a hydrocarbon fuel such as diesel or natural gas. In aspects where the prime mover **104** includes an electric motor, associated equipment such as an AC/DC converter for converting alternating current received from a power source to direct current for a direct-current electric motor. The control interface **106** can be coupled with the prime mover **104** and can have associated logic to control the rotational speed of the motor.

(20) The first control interface **106** can be mounted to the base frame **101** or to the tower **102**. The second control interface **108** can be mounted to the hydraulic power source **107**, directly to the base frame **101**, or to the tower **102**. The first control interface **106** and second control interface **108** may be embodied in the same control computer device; alternatively, may be embodied in separate devices.

(21) The control computer that includes the control interface **106** can include one or more processors, memory, and instructions stored on the memory that cause the one or more processors to receive signals from one or more sensors associated with operation of the long-stroke pumping unit **100** (e.g., a tachometer, a load cell, an accelerometer, or combinations thereof), convert the sensor signals to values associated with the measured parameter, to display and/or store the values with an associated time stamp, and to output signals for control of one or of the equipment of the long-stroke pumping unit **100**. The control computer can be networked with any sensors for control of the upstroke and downstroke of the long-stroke pumping unit **100** via wireless or wired data transmission networking (e.g., Wi-Fi, Bluetooth, NFC, ethernet cables, or combinations thereof). In aspects, the control interface **106** can include one or more virtual or physical buttons that control mechanical operation of the long-stroke pumping unit **100**. In aspects, the control computer that includes the control interface **106** does not control the hydraulic power source **107** and any of the hydraulic equipment disclosed herein (e.g., the first vertical hydraulic jack **120**, the second vertical hydraulic jack **121**, the hydraulic cylinder **122** described hereinbelow).

(22) The control computer that includes the control interface **108** can include one or more processors, memory, and instructions stored on the memory that cause the one or more processors to receive signals from one or more sensors associated with operation of the hydraulic power source **107**, the first vertical hydraulic jack **120**, the second vertical hydraulic jack **121**, and the hydraulic cylinder **122** (e.g., pressure transducers), convert the sensor signals to values associated with the measured parameter, to display and/or store the values with an associated time stamp, and to output signals for control of the first vertical hydraulic jack **120**, the second vertical hydraulic jack **121**, and the hydraulic cylinder **122**. The control computer can be networked with any sensors for control of the hydraulics of the long-stroke pumping unit **100** via wireless or wired data transmission networking (e.g., Wi-Fi, Bluetooth, NFC, ethernet cables, or combinations thereof). In aspects, the control interface **108** can include one or more virtual or physical buttons that control valve actuation in the hydraulic power source **107**, to controllably supply pressurized hydraulic fluid at various pressures for extension and retraction of the first vertical hydraulic jack **120**, the second vertical hydraulic jack **121**, and the hydraulic cylinder **122**. In some aspects, the control interface **108** can additionally include one or more virtual or physical buttons that control valve actuation in the hydraulic power source **107**, to controllably supply pressurized hydraulic fluid at various pressures for operation of the braking system **117**.

(23) In operation for pumping hydrocarbons, the control interface **106** is used to control the prime mover **104**, e.g., to rotate the drive sprocket **110** via the shaft **105**. Rotation of the drive sprocket **110** drives the chain **111** in a loop around the drive sprocket **110** and an idler sprocket of the chain idler **112**. The carriage **113** converts the movement of the chain **111** into a vertical (upward or downward) movement of the counterweight assembly **114** within the housing **109** of the tower **102**. Vertical movement (upward and downward) of the counterweight assembly **114** moves the load belt **118** to move the hanger assembly **119** upward and downward, which moves the polished rod **205**, rod string **206**, and plunger **207** upward and downward in the production tubing **203**. After downward movement (also called a down stroke) to a barrel at the bottom of the wellbore **201**, the plunger **207** is pulled upward by the rod string **206**, polished rod **205**, hanger assembly **119**, and load belt **118** to produce fluid via the hydrocarbon production line **204**.

(24) FIG. 1B illustrates that the end **103b** of the long-stroke pumping unit **100** hangs over the surface **20** of the earth when the long-stroke pumping unit **100** is in the second position; however, the foundation **10** can be configured of any size, such as having a length greater than a length of the long-stroke pumping unit **100** such that the foundation **10** is under the long-stroke pumping unit **100** when the long-stroke pumping unit **100** is in the first position and in the second position.

(25) FIG. 1B also illustrates at least one wheel **130** that is placed on the base frame **101** (e.g., on or in a wheel-well of rail **101a** of the base frame **101** illustrated in FIGS. 2A and 2B), for horizontal movement of the base frame **101** between the first position and the second position in the direction of double headed arrow A-A. While one wheel **130** is illustrated in FIG. 1B, additional wheels can be placed on the base frame **101**, such as on or the other rail **101b** of the base frame **101** illustrated in FIGS. 2A and 2B.

(26) In operation for movement of the long-stroke pumping unit **100**, the vertical hydraulic jacks disclosed herein lift the end **103a** of the long-stroke pumping unit **100** upward from the foundation **10**, wheels (e.g., such as wheel **130**) is/are placed on the base frame **101**, the vertical hydraulic jacks lower the end **103a** of the long-stroke pumping unit **100** downward toward the foundation **10** so that wheels (e.g., wheel **130**) engages with the top surface of the foundation **10**, and the hydraulic cylinder moves the long-stroke pumping unit **100** from the first position illustrated in FIG. 1A to the second position illustrated in FIG. 1B. The hydraulic power source **107** supplies the pressurized hydraulic fluid for the vertical hydraulic jacks and the hydraulic cylinder, and the hydraulic power source **107** is controlled for movement in both the vertical and horizontal directions via the control interface **108**.

(27) FIG. 2A illustrates a top view of a long-stroke pumping unit **100** in the first position, and FIG. 2B illustrates a top view of a long-stroke pumping unit **100** in the second position. FIG. 2A corresponds to the position of the long-stroke pumping unit **100** in FIG. 1A, and FIG. 2B corresponds to the position of the long-stroke pumping unit **100** in FIG. 1B. The top **115**, drum assembly **116**, braking system **117**, load belt **118**, and hanger assembly **119** of the long-stroke pumping unit **100** in FIGS. 1A and 1B are not illustrated in FIGS. 2A and 2B for clarity in viewing the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121**.

(28) FIGS. 2A and 2B illustrate that a first distance D1 between the end **103a** of the long-stroke pumping unit **100** (which is the end **103a** of the base frame **101**, and the end **103a** of the pair of parallel rails **101a** and **101b**) when the long-stroke pumping unit **100** is in the first position is less than a second distance D2 between the end **103a** of the long-stroke pumping unit **100** when the long-stroke pumping unit **100** is in the second position. It can also be seen that the wellhead **200** in FIG. 2B is more accessible with the long-stroke pumping unit **100** in the second position, compared to the location of the long-stroke pumping unit **100** in the first position in FIG. 2A.

(29) FIGS. 2A and 2B illustrate that the long-stroke pumping unit **100** additionally includes a first vertical hydraulic jack **120**, a second vertical hydraulic jack **121**, and a hydraulic cylinder **122**.

(30) The first vertical hydraulic jack **120** is coupled to the first rail **101a** of the base frame **101**, and the second vertical hydraulic jack **120** is coupled to the second rail **101b** of the base frame **101**.

While two vertical hydraulic jacks **120** and **121** are illustrated in the figures of this application, alternative embodiments can include more than two vertical hydraulic jacks **120** and **121**. For example, a third vertical hydraulic jack can be included between the vertical hydraulic jacks **120** and **121** and connected to the same hydraulic fluid loop as the first and second vertical hydraulic jacks **120** and **121**, where the third vertical hydraulic jack is mounted on the base or bottom of the tower **102**. In another example, a third vertical hydraulic jack can be connected on an opposite side of rail **101a** (a side opposite of the side where the first vertical hydraulic jack **120** is mounted) on the end **103a** of the long-stroke pumping unit **100** and a fourth vertical hydraulic jack can be connected on an opposite side of rail **101b** (a side opposite of the side where the second vertical hydraulic jack **121** is mounted) on the end **103a** of the long-stroke pumping unit **100**, where the third and fourth vertical hydraulic jacks are both connected to the same hydraulic fluid loop as the first and second vertical hydraulic jacks **120** and **121**.

(31) The hydraulic cylinder **122** has an end connected to the foundation **10** and an opposite end connected to the tower **102** (e.g., to the housing of the tower **102**). In alternative aspects, the opposite end of the hydraulic cylinder **122** can be connected to one of the rails **101a** and **101b** of the base frame **101**. While one hydraulic cylinder **122** is illustrated in FIGS. 2A and 2B, alternative embodiments include two or more hydraulic cylinders, where each cylinder has an end connected to the foundation **10** and an opposite end connected to the tower **102**, and where each cylinder is in the same hydraulic fluid circuit or loop as the hydraulic cylinder **122**.

(32) A hydraulic fluid conduit **123** is connected to the hydraulic power source **107** and the first vertical hydraulic jack **120**, the second vertical hydraulic jack **121**, or to both the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121**. While FIGS. 2A and 2B illustrate an embodiment where the first vertical hydraulic jack **120** is hydraulically connected to the second vertical hydraulic jack **121** by hydraulic fluid conduit **124**, and the hydraulic fluid conduit **123** is connected only to the second vertical hydraulic jack **121**, alternative embodiments of hydraulic connection of the vertical hydraulic jacks **120** and **121** with the hydraulic power source **107** are contemplated. For example, it is alternatively contemplated the first vertical hydraulic jack **120** can be hydraulically connected to the second vertical hydraulic jack **121** by hydraulic fluid conduit **124**, and the hydraulic fluid conduit **123** is connected to the first vertical hydraulic jack **120**. Alternatively, it is contemplated that the hydraulic fluid conduit **123** can be split into two portions, where one portion connects directly with the first vertical hydraulic jack **120** and the second portion connects directly with the second vertical hydraulic jack **121** (and there is no hydraulic fluid conduit **124** connecting the vertical hydraulic jacks **120** and **121** to one another). In aspects, the hydraulic fluid conduit **124** can be as two or more conduits that fluidly connect hydraulic fluid in the vertical hydraulic jacks **120** and **121**. All embodiments of hydraulic connection of the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to the hydraulic power source **107** supply the same pressure of hydraulic fluid to both the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** such that first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** simultaneously extend or retract at the same rate or speed (e.g., to prevent tipping of the tower **102** of the long-stroke pumping unit **100** to a side).

(33) A hydraulic fluid conduit **125** is connected to the hydraulic power source **107** and to the hydraulic cylinder **122**.

(34) When the long-stroke pumping unit **100** is in the first position as illustrated in FIG. 2A, the hydraulic cylinder **122** is in an extended position, and when the long-stroke pumping unit **100** is in the second position as illustrated in FIG. 2B, the hydraulic cylinder **122** is in a retracted position.

(35) FIG. 3 illustrates an end elevational view of the long-stroke pumping unit **100**. The wellhead **200** and wellbore **201** are not illustrated in FIG. 3 for clarity. The hanger assembly **119** is shown disconnected from the polished rod of the wellhead **200**, for viewing of the first hydraulic jack **120** and the second hydraulic jack **121**.

(36) Whereas the load belt **118** is illustrated as being thin from the side elevational views in FIGS.

1A and **1B**, the load belt **118** can be wide when viewed from the end elevational view in FIG. **3**. The load belt **118** can be seen extending around the top of the drum of the drum assembly **116**, with the braking system **117** being operably coupled with the drum of the drum assembly **116**. As viewed in FIG. **3**, the load belt **118** is retracted into and output from the tower **102** in a manner such that the load belt **118** raises and lowers to guide the long upstroke and long downstroke of the pump that is in the wellbore **201** illustrated and described in FIGS. **1A** and **1B**.

(37) The vertical hydraulic jacks **120** and **121** can be seen in FIG. **3**. The first vertical hydraulic jack **120** is coupled to the first rail **101a** of the base frame **101** via a first mounting assembly **300**, and the second vertical hydraulic jack **121** is coupled to the second rail **101b** of the base frame **101** via a second mounting assembly **350**. The hydraulic fluid conduit **124** can be seen extending between the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121**. While a single hydraulic fluid conduit **124** is illustrated in FIG. **3**, multiple hydraulic fluid conduits can be used, each being connected to both the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121**.

(38) FIG. **4A** illustrates a close-up end elevational view of the long-stroke pumping unit **100**, with the vertical hydraulic jacks **120** and **121** in a retracted position; and FIG. **4B** illustrates a close-up end elevational view of the long-stroke pumping unit **100**, with the vertical hydraulic jacks **120** and **121** in an extended position.

(39) In FIGS. **4A** and **4B**, the first vertical hydraulic jack **120** has a housing **126** and a rod **127** that is movable within the housing **126** (e.g., the rod **127** is extendable from the housing **126** and retractable into the housing **126**, between a retracted position and an extended position). An end of the rod **127** of the first vertical hydraulic jack **120** faces the foundation **10**, e.g., faces a top surface or contact surface of the foundation **10**, where a plane of the top surface or contact surface is substantially perpendicular to the direction of movement of the rod **127**.

(40) In FIGS. **4A** and **4B**, the second vertical hydraulic jack **121** has a housing **128** and a rod **129** that is movable within the housing **128** (e.g., the rod **129** is extendable from the housing **128** and retractable into the housing **128**, between a retracted position and an extended position). An end of the rod **129** of the second vertical hydraulic jack **121** faces the foundation **10**, e.g., faces a top surface or contact surface of the foundation **10**, where a plane of the top surface or contact surface is substantially perpendicular to the direction of movement of the rod **129**.

(41) The first vertical hydraulic jack **120** is coupled to the rail **101a** by the first mounting assembly **300**. The first mounting assembly **300** can include a bracket **301** connected to the rail **101a**, e.g., via welds. The first mounting assembly **300** can further include a cage **310** comprising a top portion **302**, rods **304**, and a bottom portion **305**. The cage **310** is configured to hold and contain the first vertical hydraulic jack **120** in a vertical orientation. As used herein “vertical orientation” when used to refer to the orientation of the first vertical hydraulic jack **120** means that the longitudinal movement of the rod **127** of the first vertical hydraulic jack **120** is in a vertical direction, i.e., the direction of double headed arrow B-B in FIGS. **4A** and **4B**. The top portion **302** is connected to the bracket **301** by a first connector **303** (e.g., embodied as a pin, rod, screw, or a nut and bolt). The top portion **302** is also connected to rods **304**, e.g., by welds, by threads on the end of the rods **304**, or by connectors (e.g., embodied as screws, lugs, or bolts) that secure the top portion **302** to the rods **304**. Opposite ends of the rods **304** are connected to the bottom portion **305**, e.g., by welds, by threads on the end of the rods **304**, or by connectors (e.g., embodied as screws, lugs, or bolts) that secure the bottom portion **305** to the rods **304**. The first vertical hydraulic jack **120** is contained within the cage **310**, e.g., contained within the top portion **302**, rods **304**, and bottom portion **305**. The first mounting assembly **300** can further include one or more guide members (e.g., guide member **306** being illustrated in FIGS. **4A** and **4B**). The guide member(s) **306** is connected to the rail **101a** and is configured to maintain the cage **310** in a vertical orientation (thus maintaining the first vertical hydraulic jack **120** in a vertical orientation). Each guide member **306** of the first mounting assembly **300** can be secured to cage **310**, such as to the bottom portion

305 of the cage **310** via a connector (e.g., embodied as a weld, screw, lug, or bolt).

(42) The second vertical hydraulic jack **121** is coupled to the rail **101b** by the second mounting assembly **350**. The second mounting assembly **350** can include a bracket **351** connected to the rail **101b**, e.g., via welds. The second mounting assembly **350** can further include a cage **360** comprising a top portion **352**, rods **354**, and a bottom portion **355**. The cage **360** is configured to hold and contain the second vertical hydraulic jack **121** in a vertical orientation. As used herein “vertical orientation” when used to refer to the orientation of the second vertical hydraulic jack **121** means that the longitudinal movement of the rod **129** of the second vertical hydraulic jack **121** is in a vertical direction, i.e., the direction of double headed arrow B-B in FIGS. **4A** and **4B**. The top portion **352** is connected to the bracket **351** by a first connector **353** (e.g., embodied as a pin, rod, screw, or a nut and bolt). The top portion **352** is also connected to rods **354**, e.g., by welds, by threads on the end of the rods **354**, or by connectors (e.g., embodied as screws, lugs, or bolts) that secure the top portion **352** to the rods **354**. Opposite ends of the rods **354** are connected to the bottom portion **355**, e.g., by welds, by threads on the end of the rods **354**, or by connectors (e.g., embodied as screws, lugs, or bolts) that secure the bottom portion **355** to the rods **354**. The first vertical hydraulic jack **121** is contained within the cage **360**, e.g., contained within the top portion **352**, rods **354**, and bottom portion **355**. The second mounting assembly **350** can further include one or more guide members (e.g., guide member **356** being illustrated in FIGS. **4A** and **4B**). The guide member(s) **356** is connected to the rail **101b** and is configured to maintain the cage **360** in a vertical orientation (thus maintaining the second vertical hydraulic jack **121** in a vertical orientation). Each guide member(s) **356** of the second mounting assembly **350** can be secured to cage **360**, such as to the bottom portion **355** of the cage **360** via a connector (e.g., embodied as a weld, screw, lug, or bolt).

(43) Operation can be described starting with the view in FIG. **4A**. Pressurized hydraulic fluid (e.g., a liquid primarily including mineral oil or water) can be provided to the first vertical hydraulic jack **120** and to the second vertical hydraulic jack **121** via operation of the hydraulic power source **107** via input to the control interface **108**. The pressurized hydraulic fluid can cause the rods **127** and **129** to extend out of the housings **126** and **128** to contact the top surface of the foundation **10** and then lift the end of the long-stroke pumping unit **100** vertically upward such that a space is formed between the bottom of the end of the long-stroke pumping unit **100** and the top surface of the foundation **10**, as illustrated in FIG. **4B**. A height **H1** of the space is indicated in FIG. **4B**, and the height **H1** is of a dimension sufficient for field personnel to place wheels on or in wheel wells of the rails **101a** and **101b**. Once wheels are placed on the rails **101a** and **101b**, pressurized fluid can be controllably released via input to the control interface **108** from the vertical hydraulic jacks **120** and **121** to cause the rods **127** and **129** to begin retraction back into the housings **126** and **128**, thus lowering the end of the long-stroke pumping unit **100** vertically downward toward the top surface of the foundation **10**. The rods **127** and **129** can be further retracted into the housings **126** and **128**, to a retracted position in FIG. **4A**.

(44) Guiderrails **11** and **12** are also illustrated in FIGS. **4A** and **4B**. The guiderrails **11** and **12** can be attached to the foundation **10** to prevent movement of the long-stroke pumping unit **100** beyond the guiderrails **11** and **12** (e.g., when the hydraulic cylinder **122** is actuated to move the long-stroke pumping unit **100** from the second position back to the first position). The configuration of the guiderrails **11** and **12** is not limited by the disclosure, and additional guiderrails to prevent side-to-side movement of the long-stroke pumping unit **100** can be utilized.

(45) FIG. **5** illustrates a perspective view of the second vertical hydraulic jack **121** on the end of the long-stroke pumping unit **100**. The bracket **351** of the second mounting assembly **350** can be seen connected to the rail **101b** of the base frame **101**. The cage **360** can be seen coupled to the bracket **351**, and the guide member **356** is connected to the bottom portion **355** of the cage **360** to maintain the second vertical hydraulic jack **121** in a vertical orientation.

(46) Methods

(47) A method disclosed herein involves using the hydraulic power source **107** to control, via a single control interface **108** for the hydraulic power source **107**, the vertical hydraulic jacks **120** and **121** and the hydraulic cylinder **122** for vertical and horizontal movement of the long-stroke pumping unit **100**.

(48) One method can include simultaneously actuating, with the hydraulic power source **107** when the hydraulic cylinder **122** of the long-stroke pumping unit **100** is in an extended position, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lift an end **103a** of the long-stroke pumping unit **100** upward such that a space is formed between the end **103a** of the long-stroke pumping unit **100** and the foundation **10** that is under the long-stroke pumping unit **100**; placing wheels **130** onto the long-stroke pumping unit **100** via the space; and simultaneously actuating, with the hydraulic power source **107**, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** while the wheels **130** are on the long-stroke pumping unit **100**. The method can also include, after placing the wheels **130**, actuating, with the hydraulic power source **107**, the hydraulic cylinder **122** from the extended position to a retracted position to move the long-stroke pumping unit **100** from the first position to the second position while the wheels **130** are on the long-stroke pumping unit **100**. The method can additionally include actuating, with the hydraulic power source **107**, the hydraulic cylinder **122** from the retracted position to the extended position to move the long-stroke pumping unit **100** from the second position to the first position while the wheels **130** are on the long-stroke pumping unit **100**; simultaneously actuating, with the hydraulic power source **107** while the hydraulic cylinder **122** is in the extended position, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lift the end **103a** of a long-stroke pumping unit **100** upward such that the space is again formed between the end **103a** of the long-stroke pumping unit **100** and the foundation **10** while the long-stroke pumping unit **100** is in the first position; removing the wheels **130** from the long-stroke pumping unit **100** via the space while the long-stroke pumping unit **100** is in the first position; and simultaneously actuating, with the hydraulic power source **107** while the hydraulic cylinder **122** is in the extended position, the first vertical hydraulic jack **121** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** to directly rest on the foundation **10** while the long-stroke pumping unit **100** is in the first position. The control interface **108** can control operation of the hydraulic power source **107** in all steps of the method. In aspects, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** are not removed from the long-stroke pumping unit **100** after performance of any step of the method, e.g., after simultaneously actuating the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** while the wheels **130** are on the long-stroke pumping unit **100**, or after simultaneously actuating the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** to directly rest on the foundation **10** (without wheels **130**) while the long-stroke pumping unit is in the first position.

(49) Another method can include simultaneously actuating, with the hydraulic power source **107** while a hydraulic cylinder **122** of a long-stroke pumping unit **100** is in an extended position, a first vertical hydraulic jack **120** and a second vertical hydraulic jack **121** to lift an end **103a** of the long-stroke pumping unit **100** upward such that a space is formed between the end **103a** of the long-stroke pumping unit **100** and a foundation **10** that is under the long-stroke pumping unit **100** while the long-stroke pumping unit **100** is in a first position; removing wheels **130** from the long-stroke pumping unit **100** via the space; and simultaneously actuating, with the hydraulic power source **107** while the hydraulic cylinder **122** of the long-stroke pumping unit **100** is in the extended position, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** to directly rest on the foundation **10** while the long-stroke pumping unit **100** is in the first position. The method can additionally include prior to simultaneously actuating the first vertical hydraulic jack **120** and the second vertical hydraulic jack

121 to lift the end **103a** of the long-stroke pumping unit **100**, actuating, with the hydraulic power source **107**, the hydraulic cylinder **122** from the retracted position to the extended position to move the long-stroke pumping unit **100** from a second position to the first position while the wheels **130** are on the long-stroke pumping unit **100**. The method can also include after actuating, with the hydraulic power source **107**, the hydraulic cylinder **122** from the retracted position to the extended position, simultaneously actuating, with the hydraulic power source **107** when the hydraulic cylinder **122** is in the extended position, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lift the end **103a** of the long-stroke pumping unit **100** upward such that the space is again formed between the end **103a** of the long-stroke pumping unit **100** and the foundation **10**, wherein the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** are connected to the end **103a** of the long-stroke pumping unit **100**; placing the wheels **130** onto the long-stroke pumping unit **100** via the space; and simultaneously actuating, with the hydraulic power source **107**, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **130a** of the long-stroke pumping unit **100** while the wheels **130** are on the long-stroke pumping unit **100**. The method can also include actuating, with the hydraulic power source **107**, the hydraulic cylinder **122** from the extended position to the retracted position to move the long-stroke pumping unit **100** from the first position to the second position while the wheels **130** are on the long-stroke pumping unit **100**. The control interface **108** can control operation of the hydraulic power source **107** in all steps of the method. In aspects of the method, the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** are not removed from the long-stroke pumping unit **100** after simultaneously actuating the first vertical hydraulic jack **120** and the second vertical hydraulic jack **121** to lower the end **103a** of the long-stroke pumping unit **100** to directly rest on the foundation **10** while the long-stroke pumping unit **100** is in the first position.

(50) In aspects of the methods, a first distance **D1** between the long-stroke pumping unit **100** and the wellhead **200** when the long-stroke pumping unit **100** is in the first position is less than a second distance **D2** between the long-stroke pumping unit **100** and the wellhead **200** when the long-stroke pumping unit **100** is in the second position.

Additional Description

(51) Aspect 1. A long-stroke pumping unit comprising: a base frame comprising a pair of parallel rails, wherein the base frame is configured to be positioned on a foundation that is proximate to a wellhead, wherein an end of the base frame faces the wellhead; a tower connected to each of the pair of parallel rails near the end of the base frame; a hydraulic cylinder having an end connected to the base frame or to the tower and an opposite end connected to the foundation; a first vertical hydraulic jack coupled to one of the pair of parallel rails at the end of the base frame; a second vertical hydraulic jack coupled to another of the pair of parallel rails at the end of the base frame; a hydraulic power source fluidly connected to the hydraulic cylinder, to the first vertical hydraulic jack, and to the second vertical hydraulic jack; and a control interface coupled to the hydraulic power source and configured to control the hydraulic power source for a movement of i) the hydraulic cylinder between a first extended position and a first retracted position, ii) the first vertical hydraulic jack between a second extended position and a second retracted position, and iii) the second vertical hydraulic jack between a third extended position and a third retracted position.

(52) Aspect 2. The long-stroke pumping unit of Aspect 1, wherein each of the first vertical hydraulic jack and the second vertical hydraulic jack comprises a housing and a rod movable within the housing, wherein an end of the rod of each of the first vertical hydraulic jack and the second vertical hydraulic jack faces the foundation.

(53) Aspect 3. The long-stroke pumping unit of any one of Aspects 1 to 2, being in a first position while the hydraulic cylinder is in the first extended position.

(54) Aspect 4. The long-stroke pumping unit of any one of Aspects 1 to 3, being in a second position while the hydraulic cylinder is in the first retracted position.

(55) Aspect 5. The long-stroke pumping unit of Aspect 3 or 4, where the first vertical hydraulic

jack and the second vertical hydraulic jack are connected to the end of the base frame when the long-stroke pumping unit is in the first position and when the long-stroke pumping unit is in the second position.

(56) Aspect 6. The long-stroke pumping unit of any one of Aspects 1 to 5, further comprising: a first mounting assembly connected to the one of the pair of parallel rails and to the first vertical hydraulic jack; and a second mounting assembly connected to the another of the pair of parallel rails and to the first vertical hydraulic jack.

(57) Aspect 7. The long-stroke pumping unit of Aspect 6, wherein: the first mounting assembly comprises a first bracket connected to the one of the pair of parallel rails, and a first cage connected to the first bracket, wherein the first cage is configured to hold and contain the first vertical hydraulic jack in a vertical orientation, and the second mounting assembly comprises a second bracket connected to the another of the pair of parallel rails, and a second cage connected to the second bracket, wherein the second cage is configured to hold and contain the second vertical hydraulic jack in the vertical orientation.

(58) Aspect 8. The long-stroke pumping unit of Aspect 7, wherein: the first mounting assembly further comprises a first guide member connected to the one of the pair of parallel rails and to the first cage, and the second mounting assembly further comprises a second guide member connected to the another of the pair of parallel rails and to the second cage.

(59) Aspect 9. The long-stroke pumping unit of Aspect 8, wherein: the first cage comprises a first top portion connected to the first bracket, a first plurality of rods having ends connected to the first top portion, and a first bottom portion connected to opposite ends of the first plurality of rods, wherein the first vertical hydraulic jack is contained within the first top portion, the first plurality of rods, and the first bottom portion, and the second cage comprises a second top portion connected to the second bracket, a second plurality of rods having ends connected to the second top portion, and a second bottom portion connected to opposite ends of the second plurality of rods, wherein the second vertical hydraulic jack is contained within the second top portion, the second plurality of rods, and the second bottom portion.

(60) Aspect 10. The long-stroke pumping unit of Aspect 9, wherein: the first guide member is connected to the first bottom portion of the first cage; and the second guide member is connected to the second bottom portion of the second cage.

(61) Aspect 11. A method comprising: simultaneously actuating, with a hydraulic power source when a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the long-stroke pumping unit; placing wheels onto the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit while the wheels are on the long-stroke pumping unit.

(62) Aspect 12. The method of Aspect 11, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from the extended position to a retracted position to move the long-stroke pumping unit from a first position to a second position while the wheels are on the long-stroke pumping unit, wherein a first distance between the long-stroke pumping unit and a wellhead when the long-stroke pumping unit is in the first position is less than a second distance between the long-stroke pumping unit and the wellhead when the long-stroke pumping unit is in the second position.

(63) Aspect 13. The method of Aspect 11 or 12, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to

lower the end of the long-stroke pumping unit.

(64) Aspect 14. The method of Aspect 12 or 13, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position to move the long-stroke pumping unit from the second position to the first position while the wheels are on the long-stroke pumping unit; simultaneously actuating, with the hydraulic power source while the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of a long-stroke pumping unit upward such that the space is again formed between the end of the long-stroke pumping unit and the foundation while the long-stroke pumping unit is in the first position; removing wheels from the long-stroke pumping unit via the space while the long-stroke pumping unit is in the first position; and simultaneously actuating, with the hydraulic power source while the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

(65) Aspect 15. The method of Aspect 14, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

(66) Aspect 16. A method comprising: simultaneously actuating, with a hydraulic power source while a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit while the long-stroke pumping unit is in a first position; removing wheels from the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source while the hydraulic cylinder of the long-stroke pumping unit is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

(67) Aspect 17. The method of Aspect 16, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

(68) Aspect 18. The method of Aspect 16, further comprising: prior to simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of the long-stroke pumping unit, actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position to move the long-stroke pumping unit from a second position to the first position while the wheels are on the long-stroke pumping unit, wherein a first distance between the long-stroke pumping unit and a wellhead when the long-stroke pumping unit is in the first position is less than a second distance between the long-stroke pumping unit and the wellhead when the long-stroke pumping unit is in the second position.

(69) Aspect 19. The method of Aspect 18, further comprising: after actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position, simultaneously actuating, with the hydraulic power source when the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of the long-stroke pumping unit upward such that the space is again formed between the end of the long-stroke pumping unit and the foundation, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the long-stroke pumping unit; placing the wheels onto the long-stroke pumping unit via the space; and simultaneously actuating, with the

hydraulic power source, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit while the wheels are on the long-stroke pumping unit.

(70) Aspect 20. The method of Aspect 19, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from the extended position to the retracted position to move the long-stroke pumping unit from the first position to the second position while the wheels are on the long-stroke pumping unit.

(71) Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. A long-stroke pumping unit comprising: a base frame comprising a pair of parallel rails, wherein the base frame is configured to be positioned on a foundation that is proximate to a wellhead, wherein an end of the base frame faces the wellhead; a tower connected to each of the pair of parallel rails near the end of the base frame; a hydraulic cylinder having an end connected to the base frame or to the tower and an opposite end connected to the foundation; a first vertical hydraulic jack coupled to one of the pair of parallel rails at the end of the base frame; a second vertical hydraulic jack coupled to another of the pair of parallel rails at the end of the base frame; a hydraulic power source fluidly connected to the hydraulic cylinder, to the first vertical hydraulic jack, and to the second vertical hydraulic jack; and a control interface coupled to the hydraulic power source and configured to control the hydraulic power source for a movement of i) the hydraulic cylinder between a first extended position and a first retracted position, ii) the first vertical hydraulic jack between a second extended position and a second retracted position, and iii) the second vertical hydraulic jack between a third extended position and a third retracted position.
2. The long-stroke pumping unit of claim 1, wherein each of the first vertical hydraulic jack and the second vertical hydraulic jack comprises a housing and a rod movable within the housing, wherein an end of the rod of each of the first vertical hydraulic jack and the second vertical hydraulic jack faces the foundation.
3. The long-stroke pumping unit of claim 1, being in a first position while the hydraulic cylinder is in the first extended position.
4. The long-stroke pumping unit of claim 3, being in a second position while the hydraulic cylinder is in the first retracted position.
5. The long-stroke pumping unit of claim 4, where the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the base frame when the long-stroke pumping unit is in the first position and when the long-stroke pumping unit is in the second position.
6. The long-stroke pumping unit of claim 1, further comprising: a first mounting assembly connected to the one of the pair of parallel rails and to the first vertical hydraulic jack; and a second mounting assembly connected to the another of the pair of parallel rails and to the first vertical hydraulic jack.

7. The long-stroke pumping unit of claim 6, wherein: the first mounting assembly comprises a first bracket connected to the one of the pair of parallel rails, and a first cage connected to the first bracket, wherein the first cage is configured to hold and contain the first vertical hydraulic jack in a vertical orientation, and the second mounting assembly comprises a second bracket connected to the another of the pair of parallel rails, and a second cage connected to the second bracket, wherein the second cage is configured to hold and contain the second vertical hydraulic jack in the vertical orientation.

8. The long-stroke pumping unit of claim 7, wherein: the first mounting assembly further comprises a first guide member connected to the one of the pair of parallel rails and to the first cage, and the second mounting assembly further comprises a second guide member connected to the another of the pair of parallel rails and to the second cage.

9. The long-stroke pumping unit of claim 8, wherein: the first cage comprises a first top portion connected to the first bracket, a first plurality of rods having ends connected to the first top portion, and a first bottom portion connected to opposite ends of the first plurality of rods, wherein the first vertical hydraulic jack is contained within the first top portion, the first plurality of rods, and the first bottom portion, and the second cage comprises a second top portion connected to the second bracket, a second plurality of rods having ends connected to the second top portion, and a second bottom portion connected to opposite ends of the second plurality of rods, wherein the second vertical hydraulic jack is contained within the second top portion, the second plurality of rods, and the second bottom portion.

10. The long-stroke pumping unit of claim 9, wherein: the first guide member is connected to the first bottom portion of the first cage; and the second guide member is connected to the second bottom portion of the second cage.

11. A method comprising: simultaneously actuating, with a hydraulic power source when a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the long-stroke pumping unit; placing wheels onto the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit while the wheels are on the long-stroke pumping unit.

12. The method of claim 11, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from the extended position to a retracted position to move the long-stroke pumping unit from a first position to a second position while the wheels are on the long-stroke pumping unit, wherein a first distance between the long-stroke pumping unit and a wellhead when the long-stroke pumping unit is in the first position is less than a second distance between the long-stroke pumping unit and the wellhead when the long-stroke pumping unit is in the second position.

13. The method of claim 12, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit.

14. The method of claim 13, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position to move the long-stroke pumping unit from the second position to the first position while the wheels are on the long-stroke pumping unit; simultaneously actuating, with the hydraulic power source while the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of a long-stroke pumping unit upward such that the space is again formed between the end of the long-stroke pumping unit and the foundation while the long-stroke

pumping unit is in the first position; removing wheels from the long-stroke pumping unit via the space while the long-stroke pumping unit is in the first position; and simultaneously actuating, with the hydraulic power source while the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

15. The method of claim 14, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

16. A method comprising: simultaneously actuating, with a hydraulic power source while a hydraulic cylinder of a long-stroke pumping unit is in an extended position, a first vertical hydraulic jack and a second vertical hydraulic jack to lift an end of the long-stroke pumping unit upward such that a space is formed between the end of the long-stroke pumping unit and a foundation that is under the long-stroke pumping unit while the long-stroke pumping unit is in a first position; removing wheels from the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source while the hydraulic cylinder of the long-stroke pumping unit is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

17. The method of claim 16, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are not removed from the long-stroke pumping unit after simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit to directly rest on the foundation while the long-stroke pumping unit is in the first position.

18. The method of claim 16, further comprising: prior to simultaneously actuating the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of the long-stroke pumping unit, actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position to move the long-stroke pumping unit from a second position to the first position while the wheels are on the long-stroke pumping unit, wherein a first distance between the long-stroke pumping unit and a wellhead when the long-stroke pumping unit is in the first position is less than a second distance between the long-stroke pumping unit and the wellhead when the long-stroke pumping unit is in the second position.

19. The method of claim 18, further comprising: after actuating, with the hydraulic power source, the hydraulic cylinder from a retracted position to the extended position, simultaneously actuating, with the hydraulic power source when the hydraulic cylinder is in the extended position, the first vertical hydraulic jack and the second vertical hydraulic jack to lift the end of the long-stroke pumping unit upward such that the space is again formed between the end of the long-stroke pumping unit and the foundation, wherein the first vertical hydraulic jack and the second vertical hydraulic jack are connected to the end of the long-stroke pumping unit; placing the wheels onto the long-stroke pumping unit via the space; and simultaneously actuating, with the hydraulic power source, the first vertical hydraulic jack and the second vertical hydraulic jack to lower the end of the long-stroke pumping unit while the wheels are on the long-stroke pumping unit.

20. The method of claim 19, further comprising: actuating, with the hydraulic power source, the hydraulic cylinder from the extended position to the retracted position to move the long-stroke pumping unit from the first position to the second position while the wheels are on the long-stroke pumping unit.
